SLAM Drone — Milestone Build Plan

Oct 21, 2026 · @kishan ballal

Overview

The project is built in layers: each week adds one layer on top of a working base, and nothing that already works gets broken. If week 3 fails, weeks 0–2 can still be presented.

When a milestone works:

1. **Freeze it.** Make a git commit and tag it (`v0-sim`, `v1-mapping`, and so on).
2. **Record it.** Capture a demo video and screenshots.
3. **Write it up.** Add that milestone's section to the report.
4. Give each milestone its **own launch file** (`m0.launch.py`, `m1.launch.py`, …). Later work never touches earlier launch files, so an old demo always still runs.

Milestone 0: Drone in a building with live sensors (Oct 21–25)

Build

- Ubuntu + ROS 2 Jazzy + Gazebo + PX4 setup.
- A custom indoor world: 3–4 rooms, a corridor, and a loop.
- The x500 drone with a 2D LiDAR and a depth camera.
- A Foxglove dashboard.

What you can show

- Fly the drone through the building with keyboard teleop.
- The dashboard shows the live camera, depth image, LiDAR scan and altitude.
- Talking point: "Simulated indoor UAV platform with full sensor suite, running in real time."

If stuck: fall back to the stock `gz_x500_depth` model in a default world. That's still a flying, sensing drone.

Milestone 1: Drone maps the building (Nov 2–8)

Build

- A simple **waypoint script** that sends PX4 offboard position setpoints along a fixed route. This avoids Nav2 for now.
- slam_toolbox builds the 2D map.
- RTAB-Map builds the 3D map (tuned light).

What you can show

- The drone flies a planned route while 2D and 3D maps grow live.
- Loop closure snaps the map into place.
- This alone already matches the project title, "SLAM-based drone."

If stuck: fly by teleop instead of the script. You still get the same mapping demo.

Milestone 2: GPS-denied + autonomous (Nov 9–15)

This milestone is split so each half is showable on its own.

2a: GPS-denied (do first)

- Disable GPS in EKF2.
- Feed rf2o laser odometry into PX4.
- Re-run the M1 mapping flight with no GPS.
- Demo: the same mapping run, now provably without GPS.

2b: Autonomous exploration

- Nav2 + explore_lite + the offboard bridge node.
- Demo: press start and the drone maps the whole building by itself.

If Nav2 fights you: write a simple custom frontier node instead. It finds the nearest unexplored frontier on the map and sends PX4 a position setpoint, skipping Nav2 entirely. It's less elegant, but still genuinely autonomous.

Milestone 3: Search-and-rescue mission and polish (Nov 16–21)

Each item is an independent bonus, so do them in this order and stop wherever time runs out.

1. **YOLO victim detection:** human models in the world, with detected people marked as pins on the map.

2. **nvblox GPU 3D mesh** replacing RTAB-Map, for the "looks amazing" upgrade. It's eye candy, not core, so it sits here rather than in week 2.
3. **Final Foxglove layout + demo video + report.**

If stuck: M2 is already done. M3 is pure upside.

Guaranteed demo at any point

If you get stuck in…	You can still present
Week 1	Sim drone in a building with a live sensor dashboard
Week 2	Drone mapping the building in 2D + 3D
Week 3	GPS-denied autonomous mapping drone
Nothing fails	Autonomous search-and-rescue drone with victim localization

The riskiest pieces (PX4 offboard control and the Nav2 bridge) come after there's already something to show, and each has a simpler fallback.